

Akabir Abbas

AI/ML and Automation Engineer

+92 303 3224737 | abbasakabir@gmail.com | linkedin.com/in/akabir-abbas | akabirabbas.me
Karachi, Pakistan | Open to Onsite, Hybrid, Remote, and Relocation Opportunities

Summary

AI/ML and Automation Engineer with professional experience developing machine learning, deep learning, Generative AI, LLM applications, AI agents, data pipelines, REST APIs, and business automation solutions. Skilled in Python, PyTorch, TensorFlow/Keras, scikit-learn, FastAPI, LangChain, PostgreSQL, Docker, MLflow, DVC, n8n, and CI/CD. Experienced in model training, evaluation, deployment, monitoring, Computer Vision, NLP, RAG, prompt engineering, API integration, and scalable AI systems.

Experience

AI Automation Engineer

Aug 2026 – Present

AIMS SERVICES, Karachi, Pakistan

- Build AI-powered business automations, AI agents, n8n workflows, REST API integrations, and third-party platform connections that reduce repetitive manual work.
- Analyze business processes and test, optimize, monitor, and maintain automation workflows for efficiency, scalability, and operational reliability.

Machine Learning Engineer

Jun 2026 – Aug 2026

Theta x Tech (SMC-PVT.) LTD, Karachi, Pakistan

- Developed Python-based AI/ML solutions, preprocessing pipelines, predictive models, LLM applications, and production-oriented REST APIs.
- Trained, evaluated, optimized, and integrated models while improving accuracy, inference performance, scalability, and reliability.

AI Automation Intern

Nov 2025 – Mar 2026

Zorq Studio, Remote

- Developed Python automation workflows, AI integrations, REST APIs, data-processing pipelines, validation modules, and reporting systems.
- Performed output validation, automated testing, debugging, exception handling, structured logging, and technical documentation.

Education

Bachelor of Science in Computer Science

2022 – 2026

IBA University

Academic Focus: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Database Systems, and Software Engineering.

Certifications: Machine Learning Engineer Track (DataCamp) | OpenAI API (DataCamp) | Google AI Essentials | MLOps Concepts | MLflow | Docker

Technical Skills

Machine Learning: Python, scikit-learn, Classification, Regression, Feature Engineering, Model Training, Evaluation, Hyperparameter Tuning, Predictive Modeling

Deep Learning and Computer Vision: PyTorch, TensorFlow/Keras, Neural Networks, CNNs, OpenCV, OCR, OMR, Image Processing, Transfer Learning, Model Optimization

Generative AI and Agentic AI: OpenAI API, GPT-4.1, Gemini API, LangChain, RAG Pipelines, AI Agents, Tool Calling, Prompt Engineering, Structured Outputs, Embeddings, Vector Search

Automation and Integration: n8n, Workflow Automation, Business Process Automation, REST APIs, Third-Party Integrations, JSON Processing, Celery, Apache Airflow, OAuth 2.0

Backend and Data: FastAPI, Flask, SQL, pandas, NumPy, ETL/ELT, PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Alembic

MLOps and DevOps: MLflow, DVC, Great Expectations, Model Monitoring, Drift Detection, Docker, GitHub Actions CI/CD, Git/GitHub, pytest, Linux/WSL2, Structured Logging

Projects

ModelOps Doctor

FastAPI, PostgreSQL, MLflow, DVC, Docker

- Built a full-stack MLOps platform with model registry, dataset profiling, prediction logs, metrics, drift detection, health scoring, automated diagnosis, and retraining recommendations.
- Evaluated a churn model with **0.90 accuracy**, **1.00 precision**, **0.80 recall**, and **0.889 F1-score**; added Redis, Celery, testing, monitoring, logging, and CI/CD.

Automated Bubble Sheet and Essay Evaluation System

Python, OpenCV, CNN, OCR, NLP, GPT-4.1

- Developed an AI system combining OMR grading, OCR extraction, CNN recognition, NLP assessment, GPT-assisted scoring, feedback generation, and automated reporting.
- Integrated Google Cloud Vision API and trained Computer Vision models, achieving **98.2% accuracy** on the final CNN model.

YouTube Shorts Automation Software

Python, Gemini API, FFmpeg, Edge TTS, YouTube API

- Built an end-to-end AI automation pipeline for trend discovery, script generation, footage retrieval, voice synthesis, video rendering, subtitles, thumbnails, and automated uploads.
- Integrated multiple APIs, OAuth 2.0, RSS feeds, retry queues, ffprobe validation, analytics, thread-safe storage, and PyInstaller executable packaging.